Judge vs. React: dissociating perception from timing in VLM reflex control

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2026

Abstract

A vision-language-model (VLM) agent that must time a single action in a first-person scene can fail two ways that existing game benchmarks cannot pull apart: it can misjudge the scene (perception), or judge correctly but act too late (timing). We build a minimal, deterministic first-person pit-jump task that reduces an episode to one decision, labels each decision in- or out-of-window from the sim's own dynamics, and injects a control delay D so that lateness varies independently of judgment. In the dimensionless coordinate $r = vD/W$ (distance travelled during the delay over window width), a policy that fires at the window's leading edge clears iff $r < 1$ by construction; oracle sweeps on the pit ($r^\star = 1.04$, the edge bracketed within one tick of delay at every pit width) and on a second rotational-aiming task ($r^\star = 0.98$) confirm that the apparatus reproduces this, and serve as calibration rather than as an empirical law. The main result is a **real model swept through that boundary**: a closed-loop Sonnet 4.6 agent, judging every frame itself, clears **32/32** in-window decisions below $r \approx 1$ and **0/26** above it ($r^\star = 1.05$), with nothing varied but the injected delay — the same correct decision succeeds paused and fails late. Past the boundary its *only* surviving episodes are ones that misjudged early, their perception error cancelled by the delay: the two failure axes compensate, which an aggregate score cannot see. The real judge also exposes two things a single-frame benchmark cannot: a per-frame judge is a first-passage process, so a 1–2% per-frame false-JUMP rate fires almost surely somewhere over the ∼90-frame approach unless a commit rule suppresses it; and its commit point is nearly deterministic (63/72 in-window decisions within one tick), so the real transition is a step, not a psychometric curve. On the perception axis, blind single-frame judgment at zero delay scales with model capability (Opus 90%, Sonnet 80%) and collapses to chance under fog. Measuring five real vision models' decision latency against the task's 257 ms window, **all land past the boundary** — hosted frontier models at $r \approx 4$–6, local open VLMs at $r \approx 31$–47 — so none can act inside the window in real time, yet the same decisions succeed when the world is paused. Pausable/turn-based deployment converts a slow-but-accurate VLM from failing to viable, and "VLMs are too slow to play games" is partly an artifact of the time regime they are benchmarked in.

1 Introduction

When a vision-language model plays a real-time game and walks into a pit, the post-mortem is ambiguous. Did it fail to *see* that a jump was needed (perception), or did it see correctly and *act too late* (timing)? Aggregate game benchmarks — score, win rate, levels cleared — fold these two failures into one number, so "the VLM is bad at this game" can mean either, and usually means

some unknown mix. The two have opposite fixes: a perception failure wants a better model; a timing failure wants a slower clock. Conflating them is why "VLMs are too slow to play games" is repeated as though it were a single fact.

Our move is isolation. We reduce an episode to *one* correct decision — a first-person pit the player auto-walks toward, where the only choice is *when* to jump — so that on any given run exactly one of the two failures can occur, and we can say which. The apparatus is deterministic and headless, and it injects a control delay D (a decision made at tick t is applied at $t{+}D$), which lets us vary *lateness* completely independently of *judgment quality*: a perfect-judgment oracle still fails if its action lands too late.

The headline result (§4.4) is a closed-loop Sonnet 4.6 agent, judging every frame itself, swept through the boundary by injected delay: it clears 32/32 in-window decisions below $r \approx 1$ and 0/26 above it, and its only clears past the boundary are *delay-rescued misjudgments*. Around it:

- **Calibration (§4.1–4.3).** For a policy that fires at the window's leading edge, clearing iff $r < 1$ is arithmetic, not a discovery. Oracle sweeps on the pit ($r^\star = 1.04$ on a 1-tick delay grid that brackets the edge at every width) and on a rotational aiming task ($r^\star = 0.98$) check that the sim's dynamics, the in-window label, and r reproduce it without leaking tolerance, and the oracle's same-decision flip previews the real-model result on a perfect judge.
- **Two things the real judge revealed (§4.5, §4.6).** A per-frame judge is a first-passage process: a false-JUMP rate that rounds to zero per frame fires almost surely per approach unless a commit rule suppresses it. And the judge's commit point is nearly deterministic, so its transition is a step rather than a curve.
- **Perception axis (§4.7).** Blind single-frame judgment at zero delay scales with model capability and collapses to chance under fog — only in the models that read depth.
- **Latency axis (§4.8).** Five real vision models' decision latencies placed on the same axis: at this sim speed all are past the boundary, hosted frontier models at $r \approx 4$–6 and local open VLMs at $r \approx 31$–47.
- **Control (Appendix A).** Action-repeat adds polling delay and shifts the transition to lower explicit r; it is a cost lever, not a latency fix.

2 Related work

Five works are closest; we differentiate from each.

- **OmniGameArena** (Lin et al., 2026, arXiv:2606.09826) — has paused (PDQ) vs. latency-controlled real-time (LCRT) clock modes, but only as **aggregate whole-game** comparisons; no per-decision in-window label, no single-decision flip. We add both, and the r coordinate that makes latency measurements comparable across tasks.
- **VideoGameBench** (Zhang et al., 2025, arXiv:2505.18134) — "Lite" (paused) vs. real-time; names the stale-action mechanism but leaves perception confounded ("it's not just latency"). We resolve exactly that confound by isolating one decision.
- **"Win Fast or Lose Slow"** (Kang et al., NeurIPS 2025, arXiv:2505.19481) — owns the latency–quality tradeoff, but its latency lever is quantization/model-size, which *also* degrades quality, and its ≈200 ms point is an environment action-rate cap, not a task-window threshold.

Our sweep varies latency **independently** of the model — the apparatus injects control delay D with the policy fixed.

- **BLINK** (Fu et al., ECCV 2024, arXiv:2404.12390) — canonical "MLLMs fail at depth," but static MCQ. Ours is interactive, parametric, oracle-normalized.
- **Ramstedt & Pal, "Real-Time RL"** (NeurIPS 2019, arXiv:1911.04448) [+ "At Human Speed," arXiv:1810.07286] — the RL parent of the real-time/delay result (monotone degradation for learned policies). We express the same delay cost as r for *pausable VLM agents*, show a real VLM crossing it, and show turn-based deployment stepping back over it.

We explicitly do **not** claim to be first to study the latency–quality tradeoff (Win Fast) nor first to isolate timing from competence in the aggregate (Real-Time Deadlines, arXiv:2601.13206; OmniGameArena). Our claim is the single-decision dissociation on a real model, the compensating-axes observation past the boundary, and the quality-controlled r coordinate that makes the latency measurements interpretable.

3 Task and apparatus

A first-person raycaster. The player auto-walks toward a pit; the only decision is *when* to jump. Knobs: pit width (sets the action-window W), fog/visibility (perception), approach speed v, seed. The sim is deterministic and headless; control delay D (ticks) is **injected** — a decision made at tick t is applied at $t{+}D$ — so D is an independent variable decoupled from wall-clock. Real per-model latency L is measured separately (timed calls) and mapped to the r axis via L/tick. An oracle policy (true pit distance) is the perfect-judgment ceiling.

Ground-truth label. `in_window` = whether a delay-0 takeoff at the decision position clears, computed from the sim's *own* dynamics (not an analytic approximation), so `in_window` exactly predicts `cleared` at $D{=}0$. The action window is $W = w_{hi} - w_{lo}$ with w_{lo}, w_{hi} derived by simulating takeoffs; for pit widths beyond the jump reach the window is empty ($W \leq 0$, unclearable) and those configs are excluded from fits (reported, not silently dropped).

$$r = \frac{v \cdot D}{W} \text{ (oracle sweep)} \qquad \text{or} \qquad r = \frac{v \cdot (L/\text{tick})}{W} \text{ (real-model placement).}$$

What r does and does not tell you. A policy that commits at the window's leading edge w_{lo} has its delayed takeoff land at $w_{lo} + vD$, which clears iff $vD < W$, i.e. iff $r < 1$. The location of the boundary is therefore fixed by where the policy fires, not discovered by sweeping; a policy that fired at the window's center would place it at $r = 0.5$. r is the coordinate in which that statement is exact and task-independent, which is what makes a measured latency L interpretable against any window. The oracle sweeps in §4.1–4.3 are calibration: they check that the dynamics, the label, and r reproduce the boundary with no leaked tolerance. The question with empirical content — where a *real* judge's commit point falls, and whether its outcome flips at that boundary with nothing else varied — is §4.4.

4 Results

4.1 Calibration: the oracle boundary

The oracle fires at the first in-window position, so a delayed takeoff lands at $w_{lo} + vD$ and clears iff $r < 1$; the sweep's job is to confirm that the apparatus reproduces this. It does. The default grid (4 pit widths $\times$ 7 delays, $D \in \{0, 4, \ldots, 24\}$) has 28 unique configurations. The 1680 committed rows replicate each 60$\times$ across fog and seed, which the oracle ignores (every replicate is identical), and the 7 configurations at pit width 4 have an empty window ($W \leq 0$, unclearable) and are excluded. On the remaining 21 configurations the pooled logistic fit gives $r^\star = 1.17$, but that number is a grid artifact: the last clear is at $r = 1.04$ and the first failure at $r = 1.30$, both at pit width 2, and 1.17 is the logistic midpoint of a 4-tick gap. The other widths do not straddle the edge within this grid — width 1 ($W = 2.08$) never exceeds $r = 0.81$ and always clears, and width 3 ($W = 0.08$) is already at $r = 3.5$ at the first nonzero delay and always fails.

A 1-tick delay grid ($D \in \{0, \ldots, 40\}$, widths 1–3, one seed, fog 0; 123 configurations) removes the artifact. Each width brackets the edge within a single tick of delay: width 1 between $r = 1.01$ ($D = 30$) and 1.04 ($D = 31$), width 2 between 1.04 ($D = 16$) and 1.10 ($D = 17$), and width 3 between 0.89 ($D = 1$) and 1.77 ($D = 2$), its window being only 1.1 ticks wide. Every width is a clean step (every delay below its edge clears, every delay at or above it fails), and the pooled logistic fit gives $r^\star = 1.04$ (Fig. 1). This is the calibration number used below. The clears just past $r = 1$ are physical: a jump that fires on the same tick can take off from just past the near edge, extending tolerance a fraction of a tick beyond the nominal window. The transition is a step because the oracle is deterministic.

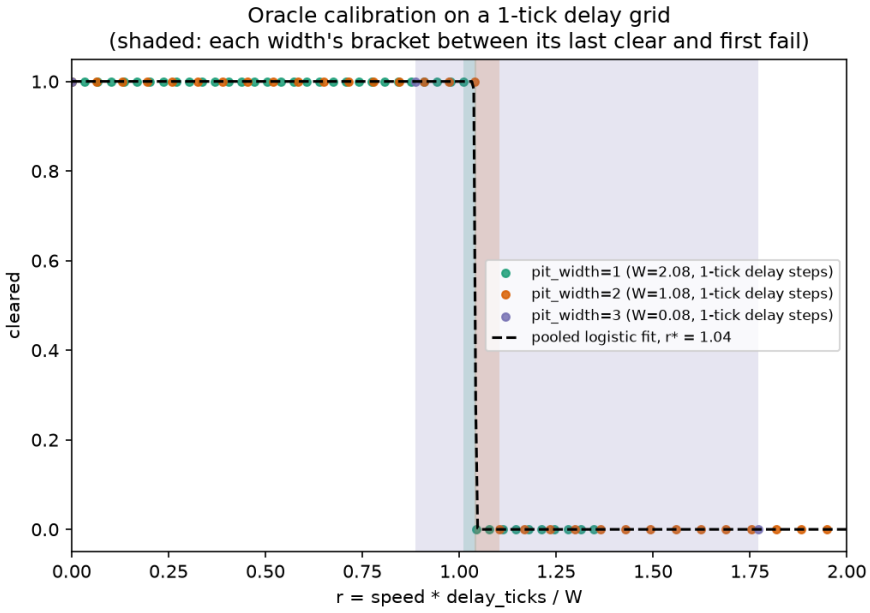


Figure 1: Calibration. Oracle clear/fail vs. r on a 1-tick delay grid; each width's shaded bracket spans its last clear to its first fail, and the pooled logistic fit gives $r^\star = 1.04$.

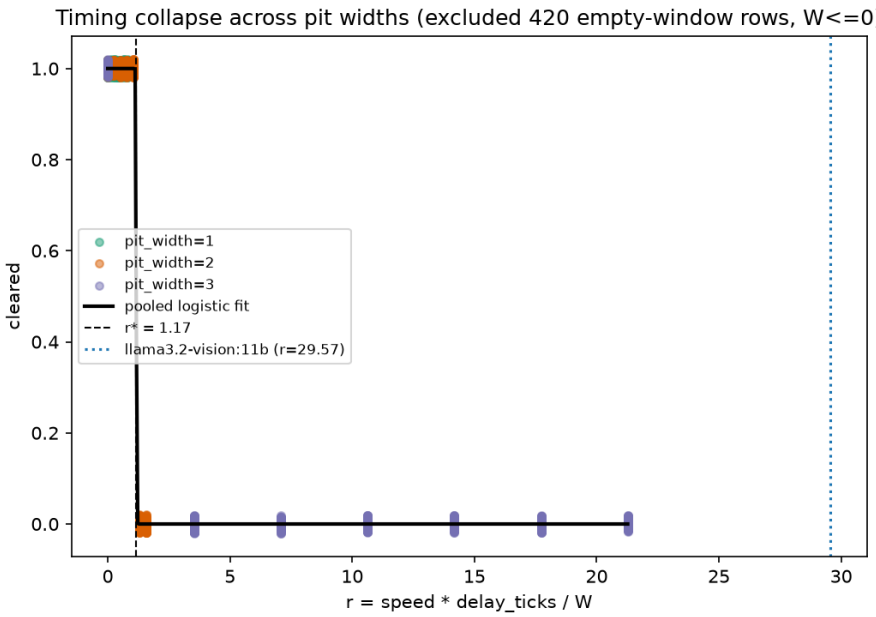


Figure 2: The default oracle grid (D in steps of 4; 21 unique configurations, each replicated 60× across fog/seed in the plotted rows). Its pooled fit of $r^\star = 1.17$ is the midpoint of a 4-tick gap at pit width 2; the 1-tick grid above supersedes it.

4.2 Same-decision flip on the oracle

Restricting to oracle-correct decisions (`in_window` = 1; the oracle's commit position does not depend on D), the **identical** decision clears 3/3 unique configurations at $D = 0$ and 0/8 once $r \geq 1.30$. The one in-window decision past $r = 1$ that clears is the $r = 1.04$ same-tick case above. Only execution timing changed. We do not summarize this as a pooled probability: six of the 9 configurations with $r > 1$ are at pit width 3, where r runs from 3.5 to 21, so a rate over them would describe the delay grid rather than the task. The flip on a real judge, with decisions on both sides of the edge, is §4.4.

4.3 The same calibration on a second task

Is the apparatus's behaviour specific to jump-timing? We built a second task in a different modality: a **sweeping-aim** task. The player rotates in place at constant angular speed ω; a target subtends an angular window $[\theta - h, \theta + h]$ (width $2h$); the single decision is *fire*, and a control delay D rotates the aim ωD further before the shot lands — overshoot if it exceeds the window. By construction the dimensionless cost is $r = \omega D/(2h)$, the exact angular analog of the pit's vD/W. The aim episode is pure kinematics and emits the same schema, so the *identical* fit machinery applies.

Because the aim oracle also fires at the leading edge, a hit iff $r < 1$ is again arithmetic; the test is whether an independent implementation of the kinematics and the label reproduces it. It does: across 205 unique configurations at five target widths (4.6°–13.8°; one seed, the aim oracle being deterministic) the pooled logistic fit gives $r^\star = 0.98$, and at every width the last hit lies at $r = 0.87$–0.96 and the first miss at exactly $r = 1.00$ (Fig. 4). The offset from the pit's 1.04 is the oracles' firing phase: the pit's takeoff can extend just *past* the near edge (a hair above 1), while the aim fires just *inside* the near edge, forfeiting a sub-tick of tolerance (a hair below 1). Two tasks agreeing on

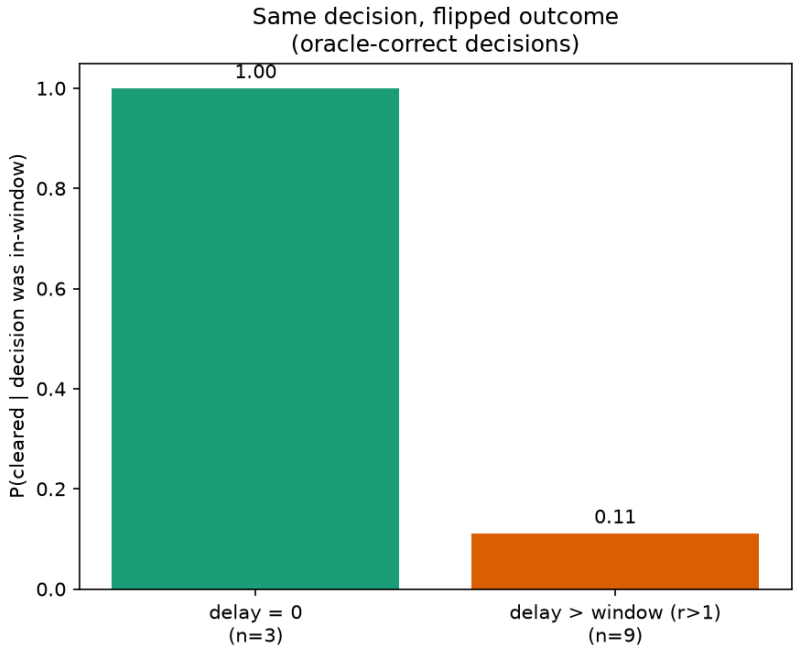


Figure 3: Oracle-correct decisions: 3/3 clear at $D = 0$; of 9 unique configurations past $r = 1$, only the $r = 1.04$ same-tick case clears (0/8 at $r \geq 1.30$).

$r \approx 1$ shows that r is the right coordinate for both and that neither implementation leaks tolerance. It is not evidence of an empirical law: a policy that fired at the window's center would place its boundary at $r = 0.5$ in the same apparatus. What the next section asks is where a real judge's commit point actually falls.

4.4 A real model through the boundary

The oracle sections calibrate the boundary; here a real model is swept *through* it with its own judgment in the loop. Because the apparatus decouples sim time from wall-clock, the model's (slow, variable) API latency is free, and r is engineered exactly as in §4.1: Sonnet 4.6 judges *every grounded frame* of a closed-loop blocking episode (pit width 2, $v = 0.07$, fog 0, seeds 1–6), its committed jump is applied D ticks later, and sweeping $D \in \{0, \ldots, 31\}$ sweeps $r = vD/W$ from 0 to 2, densely around $r = 1$ (72 episodes; 6,612 vision calls).

Judge protocol. The episode prompt used for the latency battery (§4.8) turns out never to elicit JUMP near the window from *any* hosted model: it describes the pit as "the black band across the floor ahead," which stops matching the close-range view — inside the window the pit is a black region filling the bottom of the frame, and the visible checkered floor is the pit's *far* side. The judge used here instead answers a literal perceptual question ("what touches the bottom border of the image: tiles or black?") at high reasoning effort, and commits only on two consecutive JUMP answers; §4.5 explains why that commit rule is forced rather than cosmetic. We validated the final judge on all 108 approach frames × 3 samples (Fig. 6, left): zero JUMPs on the first 79 frames, isolated JUMPs at $x = 8.10$ (1/3) and 8.24 (2/3), and unanimous JUMP from $x = 8.87$ on. This judge was developed against, and validated on, the same renderer, pit width, and fog level the sweep uses; it is a judge tuned to this scene, and we make no claim that it transfers. The bottom-edge

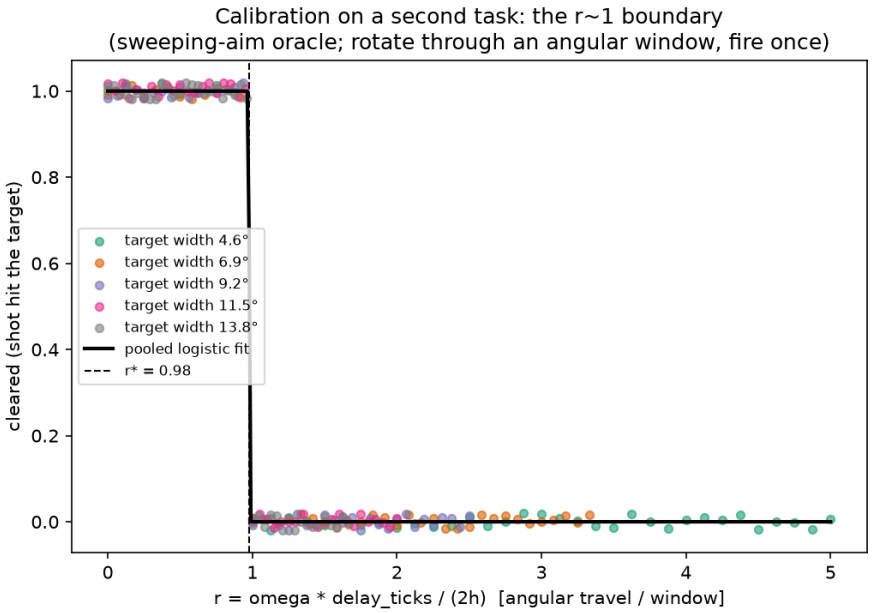


Figure 4: Calibration on a second task. The rotational aiming oracle reproduces the $r \approx 1$ boundary ($r^\star = 0.98$; 205 configurations, 5 target widths).

cue is nearly ideal in this renderer: the pit's black region first touches the frame's bottom edge at $x = 9.01$, one tick after the window opens ($w_{lo} = 8.92$).

The boundary holds for a real judge. Episodes whose decision lands in the window clear **32/32 through** $r = 0.97$, 4/5 at $r = 1.04$, and **0/26 from** $r = 1.10$ **on**; the same logistic fit as §4.1 gives $r^\star = 1.05$, coinciding with the oracle's width-2 bracket $[1.04, 1.10]$ on the 1-tick grid: the real judge's edge sits where the perfect judge's does, same-tick tolerance included (Fig. 5). The transition is nearly as sharp as the oracle's because the model's commit position is almost deterministic (§4.6); given that commit point, the *location* of $r^\star$ follows from the kinematics exactly as in §4.1. What this sweep establishes that the oracle cannot is the dissociation itself on a real judge: the same model, making the same frame-by-frame decisions, clears every in-window decision below the boundary and none above it, with nothing varied but D.

Delay-rescued misjudgments. 9/72 episodes commit early ($x = 8.10$–8.24, out of window — a perception error; these are excluded from the timing fit above). At $D \leq 8$ every such episode fails, landing short. In the band $D = 15$–24 ($r = 0.97$–1.56), *every* early episode **clears** (5/5): the delay pushes the too-early takeoff into the window. Past $r \approx 1.1$ the *only* surviving episodes are the misjudged ones; by $r = 2.01$ the delay overshoots even the early takeoff and the rescue band closes. The two failure axes do not merely dissociate — on opposite sides of the boundary they *compensate*, which is only visible because the apparatus labels each decision in- or out-of-window.

4.5 A frame-by-frame judge is a first-passage process

The commit rule above is not a prompt-engineering detail; the structure of the task forces it. A frame-by-frame controller is consulted on every approach frame, and the episode is decided by the *first* JUMP it commits to, so any per-frame false-positive rate compounds over the approach. With

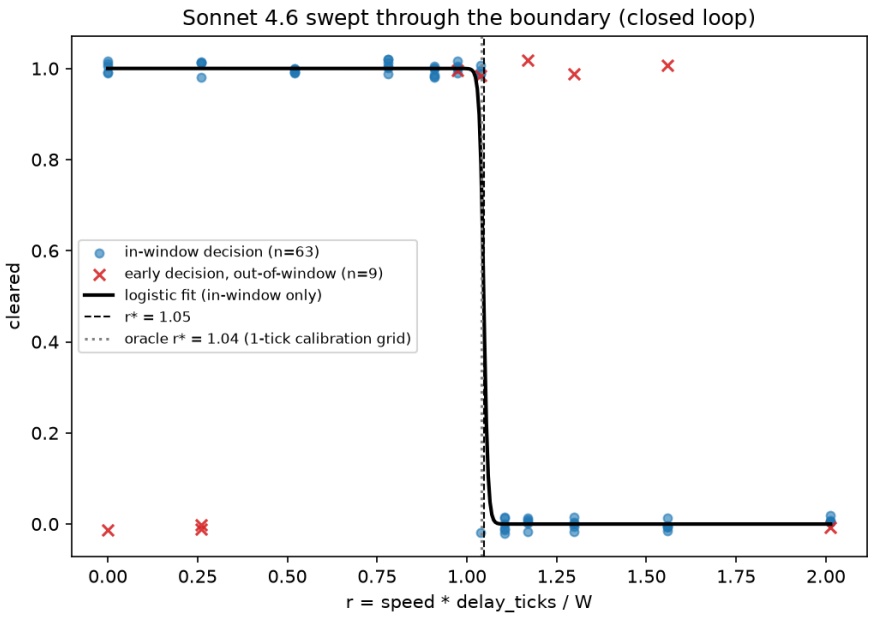


Figure 5: Sonnet 4.6 swept through the boundary in closed loop ($r^\star = 1.05$, in-window decisions). Crosses are out-of-window (early) decisions: they fail below the boundary and are the only survivors past it (delay-rescued perception errors).

a per-frame false-JUMP rate p over the $N \approx 90$ frames before the window, a single-JUMP commit rule fires somewhere early with probability $1-(1-p)^N$: **60% at $p = 1\%$ and 84% at $p = 2\%$**, the rates we measured for earlier prompt versions far from the pit. Under the same independent-error assumption the two-consecutive rule fires with probability 0.9% and 3.4% respectively (Fig. 6, right; exact two-state chain). A single-frame accuracy battery — the standard way to evaluate a VLM judge — scores a 1% false-positive rate as 99% accuracy and cannot see this. The final judge's own battery (Fig. 6, left) shows the residual: its only pre-window JUMPs are in the tile before the window ($x = 8.10$–8.24), and committing there requires two consecutive misfires, which is what the sweep's 9/72 early episodes are. Its unanimous JUMP from $x = 8.87$, one frame before the window opens, is converted by the commit rule into a commit at 8.94, inside the window. The residual errors are not independent across frames — they cluster where the floor sliver below the pit becomes thin — so the independent-error curves are an idealization; the mechanism is not. Any deployment of a VLM as a per-frame trigger needs a commit rule or an equivalent, and needs to be evaluated per episode, not per frame.

4.6 The commit point is nearly deterministic

The sweep's in-window decisions commit at $x = 8.94$ in 51/72 episodes and at $x = 9.01$ in 12 — within one tick of each other across 63 episodes and six seeds, with the judge's own sampling stochasticity (adaptive thinking at high effort) in the loop. We expected a noisy psychometric curve and got a near-step. Two readings. The optimistic one: on a rendered depth cue with a pixel-sharp onset, a frontier VLM's judgment is effectively deterministic at the tick scale, so in a deployment like this one timing failures are failures of latency, not of judgment variance — the clean separation this paper's framing assumes. The cautious one: the judge was tuned to this cue, and its one-tick lead (answering "black" at 8.94, before the region literally reaches the bottom edge at 9.01) shows

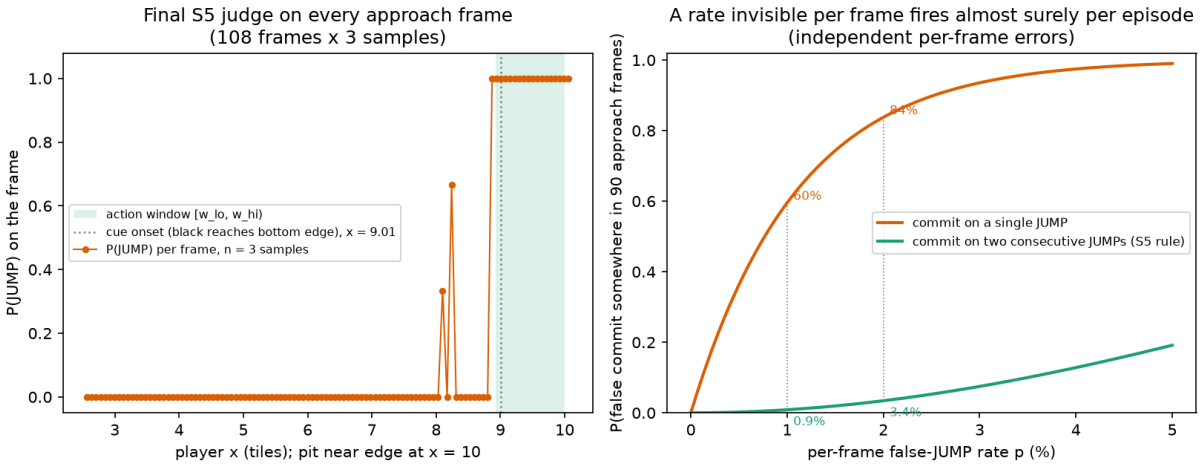


Figure 6: Left: the final judge's per-frame JUMP rate on every approach frame (108 frames × 3 samples) against the window. Right: the probability that a per-frame false-JUMP rate fires at least once over the 90 approach frames, for a single-JUMP commit rule and for the two-consecutive rule, assuming independent per-frame errors.

it is reading a thinning sliver of floor rather than the edge itself. A cue without a sharp onset would presumably spread the commit point, and that spread is the quantity a real-time deployment needs to know, because it sets how much of the window is left after the commit. Measuring it on weaker cues (fog) and weaker judges is the obvious next experiment.

4.7 Perception axis

A capable vision model *can* judge the jump from a single frame, and fog collapses it. We rendered standing decision frames across approach positions at two visibility levels and had **four blind vision models** — three Claude models (Haiku 4.5, Sonnet, Opus) and one local open VLM (`qwen2.5-vl:7b` via Ollama) — each shown only the image (no coordinates, window, or labels; identical seed-42 shuffled frame set) call JUMP or WAIT on each; we score each call against the ground-truth window. Judgment accuracy at zero delay ($n = 10$ frames per model per condition):

Table 1: Blind single-frame judgment accuracy at zero delay ($n = 10$ frames per model per condition; exact counts shown).

Model	Clear (fog=0)	Foggy (fog=1)	Fog effect
Opus	**90%** (9/10)	50% (5/10, chance)	−**40** pts
Sonnet	80% (8/10)	50% (5/10, chance)	−**30** pts
Qwen2.5-VL 7B (local)	60% (6/10)	60% (6/10)	0
Haiku 4.5	40% (4/10)	40% (4/10)	0

Two things fall out. **(1) With a clear depth cue, judgment scales with model capability** — Opus 90%, Sonnet 80%, Qwen 60%, Haiku 40%. **(2) The fog collapse appears only in the models that actually read depth.** Opus and Sonnet drop to chance under fog, and each **missed every in-window frame** (0/5): they reported no distinguishable pit band even up close, never committed, and would fall. The two weak judges show **no fog effect at all** — but for

opposite-looking reasons that are really the same reason: Haiku sits at floor (it fell back on a frame-order heuristic the shuffle nullifies), and Qwen is strongly *JUMP-biased* (it answered JUMP on 18/20 frames, clearing all in-window frames but also most out-of-window ones). Neither is using the depth cue in the first place, so neither has any perception to lose when fog removes it. The fog collapse is a **signature of genuine depth reading**, present exactly where clear-condition accuracy is high. At $n = 10$ frames per cell the per-model contrasts are directional rather than individually significant (Fisher exact, clear vs. fog: Opus $p = 0.14$, Sonnet $p = 0.35$; pooled Opus+Sonnet 17/20 vs. 10/20, $p = 0.04$) — it is the coherence of the pattern across four models, not any single cell, that carries the claim.

This is the perception half of the dissociation: a model that judges well with a clear depth cue drops to chance when fog obscures the pit's distance — **at zero delay**, i.e. independent of the timing axis. Together, perception and timing are separable failure axes on one task. (Caveat: the Claude judges run via a subagent path, not a controlled API benchmark; the local VLM is a real keyless run. Judges are blinded to coordinates and labels but not to the task.)

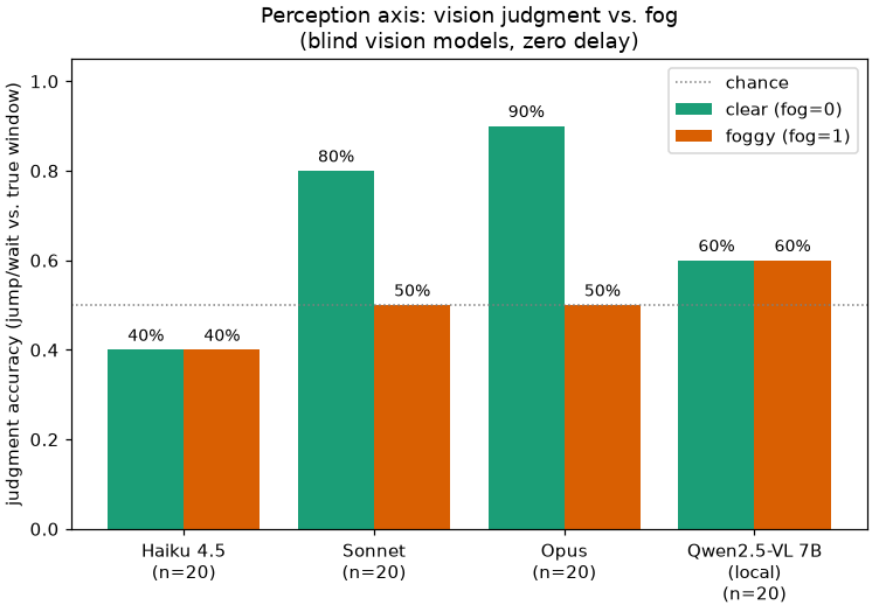


Figure 7: Perception axis. Fog collapses judgment to chance only for the depth-reading models (Opus, Sonnet); JUMP-biased/floor judges show no fog effect.

4.8 Placing real models' latency on the axis

$r = L/T_{\text{window}}$ places a measured decision latency on the same axis. At pit width 2 and $v = 0.07$ tiles/tick the window is open for $T_{\text{window}} = (W/v) \cdot \text{tick} = 257\,\text{ms}$ (tick $= 1/60\,\text{s}$), so the calibrated boundary corresponds to $r^\star \cdot T_{\text{window}} \approx 270\,\text{ms}$. We timed five vision models on the same 20-frame battery, in each model's fastest single-call configuration (Table 2, Fig. 8): hosted frontier models answer in 1–1.5 s ($r \approx 4$–6) and local open VLMs in 8–12 s ($r \approx 31$–47). None is inside the window at this speed. The multiples are a property of the speed we chose — T_{window} scales as $1/v$, and a world slowed 4× puts the hosted models at the boundary — which is the point: r is the quantity to report, and a deployment moves it by pausing or slowing the world as readily as by cutting latency. Two caveats on the measurement. These are time-to-answer numbers, not judgments: in the fast

configuration the hosted models answered WAIT on nearly every frame (Haiku and Sonnet 20/20, Opus 18/20) and the local models JUMP on nearly every frame; a configuration that discriminates (the §4.4 judge, ∼5 s/call) is slower, so these r values are lower bounds. And each local model's first call is an order of magnitude faster than the rest (0.4–0.6 s against 8–12 s), consistent with a warm cache; the medians are unaffected.

Table 2: Measured per-model decision latency (median of $n = 20$ single-frame calls, fastest configuration) against the 257 ms window at pit width 2.

Model	Median L	$r = L/\text{window}$
Haiku 4.5 (hosted API)	0.95 s	≈ 4
Sonnet 4.6 (hosted API)	1.45 s	≈ 6
Opus 4.8 (hosted API)	1.53 s	≈ 6
Qwen2.5-VL 7B (local)	7.9 s	≈ 31
llama3.2-vision 11B (local)	12.1 s	≈ 47

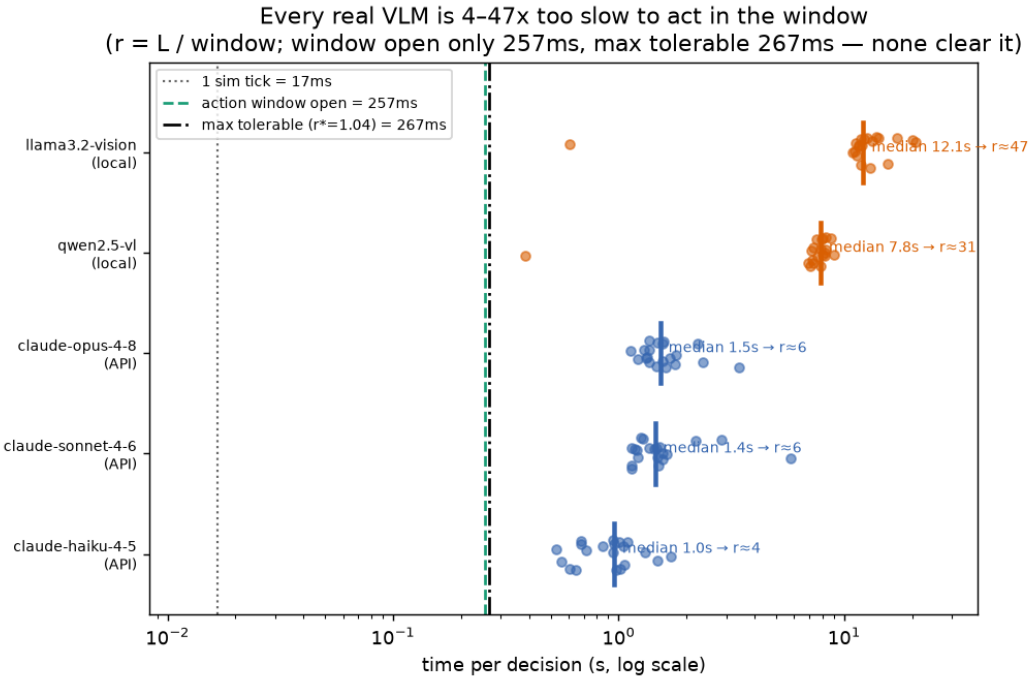


Figure 8: Real per-model decision latency vs. the time budget at this sim speed. Hosted Haiku/Sonnet/Opus ($r \approx 4$–6) and local Qwen/llama3.2-vision ($r \approx 31/47$); all past the ≈270 ms budget.

5 Discussion

The two failure axes have opposite fixes, and the framing makes the requirement quantitative. A perception failure wants a better model: no amount of extra time helps a judge that reports no distinguishable pit band under fog. A timing failure wants a slower clock: a judge that commits a fraction δ of the window past its leading edge is viable exactly when $r = L/T_{\text{window}} < 1-\delta$ (plus the

same-tick tolerance). For the oracle and for the Sonnet judge of §4.4, $\delta \approx 0.02$, which is why both sit at $r^\star \approx 1$; a judge that committed mid-window would have $r^\star = 0.5$ in the same apparatus, and the spread of its commit point (§4.6) would turn the step into a curve. A deployment can cross the boundary by pausing the world ($r \to 0$), slowing it (raising T_{window}), or cutting latency — and at this sim speed the measured field needs 4–47× on the latter two. §4.4 is the direct demonstration: the same model, same judgment, clears 32/32 below the boundary and 0/26 above it, with nothing varied but r. The practical reading is that pausable or turn-based deployment converts every slow-but-accurate model we measured from failing to viable, with no change to the model; conversely, a real-time benchmark score is a product of judgment *and* time regime, and "VLMs are too slow to play games" is partly a fact about the clock they are handed. The delay-rescued misjudgments add a sharper corollary: past the boundary, scores can even *reward* bad perception — the axes compensate, and an aggregate benchmark cannot tell. And the first-passage result (§4.5) says that even below the boundary, a per-frame judge has to be evaluated per episode: a false-positive rate that rounds to zero per frame does not round to zero per approach.

6 Limitations

- **The real-model sweep (§4.4) is one model at one pit width, at fog 0.** Its judge uses a revised prompt (the latency-battery prompt elicits no JUMPs near the window from any hosted model) plus a two-consecutive-JUMP commit rule, developed against and validated on the same 108 approach frames the sweep draws from — a judge tuned to this scene, and a documented apparatus change, not the shipped default. Its r axis is engineered by injected delay with the model's judgment in the loop; the model's own wall-clock latency (∼5 s/call at high effort) remains far past $r^\star$, so a wall-clock-real sweep is still impossible for the measured field — that is §4.8's point, not a gap. The anticipated *noisy* psychometric curve did not materialize — this judge's decision variance is sub-tick — so mapping a genuinely noisy transition needs a weaker cue or a weaker judge.
- **The location $r \approx 1$ is definitional for leading-edge policies.** The agreement between the pit and aiming oracles (§4.1, §4.3) checks the apparatus; it is not evidence that real agents on other tasks share a boundary. Whether real judges on richer, multi-decision tasks have a stable commit point — the quantity that would make r predictive for them — is untested; §4.6 shows one judge on one scene does.
- **The first-passage curves assume independent per-frame errors.** The final judge's residual errors cluster in the tile before the window, so the curves in Fig. 6 are an idealization of the mechanism, and the 1–2% per-frame rates they are evaluated at come from earlier prompt versions, not from the final judge (whose far-field rate on the battery was zero in 237 samples).
- **The perception axis** mixes three Claude judges (run via a subagent path, not a controlled API benchmark) with one real keyless local VLM (`qwen2.5-vl:7b`), at $n = 10$ frames per condition — large enough for the cross-model pattern, too small for per-model significance (Fisher exact $p = 0.14$–0.35). The open VLMs are weak/biased judges — Qwen is JUMP-biased (60% clear, no fog effect) and `llama3.2-vision:11b` is always-JUMP (contributes latency only, not a perception point). A controlled, latency-timed multi-provider API sweep still needs a key.
- **The latency battery measures time-to-answer, not judgment, at one arbitrary sim**

speed. In the fast configuration the hosted models answered WAIT on nearly every frame; a configuration that discriminates is slower, so the measured r values are lower bounds. The "4–47×" multiples scale with $1/v$ and would halve at half the approach speed; r is the portable number. Latencies are network- and hardware-dependent, but they are real decision latencies, not stand-ins.

- **No human baseline** — the oracle stands in as the perfect-judgment ceiling. Human psychophysics is future work.

7 Future work

Extending the real-model sweep to more models, pit widths, and fog levels, and measuring the commit-point spread on a cue without a sharp onset (the quantity that would turn the step into a psychometric curve); a multi-provider latency panel (Gemini/GPT + more open VLMs); a human psychophysics arm; and multi-decision tasks, to test whether real judges have a stable enough commit point for r to stay predictive beyond single reflexes.

Reproducibility

Code, data, and figure-generating scripts are at `https://github.com/star7js/judge-vs-react`, under `eval/pit/` (analysis and sweeps) and `raycaster/` (the deterministic sim, including `--task aim` for the second task). The perception judgments (`perception_judgments.jsonl`, 80 rows across 4 models), per-decision latencies (`vlm_latency.jsonl`, 100 rows across 5 models), the oracle sweeps (default grid and the 1-tick `oracle_edge_sweep.jsonl`), the aim sweep, and the real-model sweep (`eval/pit/realmodel/`: driver, revised judge, 108-frame battery validation, and the 72-episode JSONL) are committed. The local-VLM paths (Ollama) and the Claude subagent judges run without an API key; the hosted-API latency numbers and the real-model sweep required one.

A Control: action-repeat does not rescue

Consulting the policy only every k ticks and holding its last decision is a common cost lever. It adds up to $k-1$ ticks of polling delay on top of the explicit D, so it must shift the transition to *lower* explicit r, and it does. Measured $r^\star$ vs. k at pit width 2 ($D \in \{0, 2, \ldots, 40\}$, one seed): k=1 → 1.10, k=2, 4 → 0.97, k=8, 16 → 0.71; the ties are grid quantization (a 2-tick delay step is 0.13 in r at this width). This is a control, confirming that the apparatus charges polling delay the same as injected delay. Action-repeat is a compute/cost lever, not a latency fix.

References

[1] M. Lin et al. *OmniGameArena: A Unified UE5 Benchmark for VLM Game Agents with Improvement Dynamics.* arXiv:2606.09826, 2026.

[2] A. L. Zhang, T. L. Griffiths, K. Narasimhan, O. Press. *VideoGameBench: Can Vision-Language Models complete popular video games?* arXiv:2505.18134, 2025.

[3] H. Kang et al. *Win Fast or Lose Slow: Balancing Speed and Accuracy in Latency-Sensitive Decisions of LLMs.* NeurIPS 2025, arXiv:2505.19481.

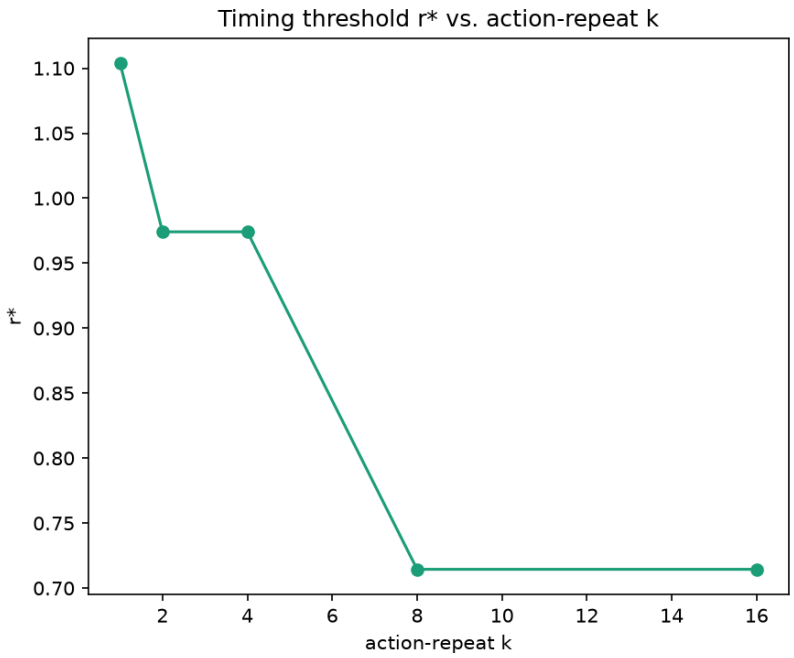


Figure 9: Action-repeat does not rescue the transition; larger k fails at lower explicit r.

[4] X. Fu et al. *BLINK: Multimodal Large Language Models Can See but Not Perceive*. ECCV 2024, arXiv:2404.12390.

[5] S. Ramstedt, C. Pal. *Real-Time Reinforcement Learning*. NeurIPS 2019, arXiv:1911.04448.

[6] V. Firoiu, T. Ju, J. Tenenbaum. *At Human Speed: Deep Reinforcement Learning with Action Delay*. arXiv:1810.07286, 2018.

[7] N. K. R. Sehgal, S. C. Guntuku, L. Ungar. *Real-Time Deadlines Reveal Temporal Awareness Failures in LLM Strategic Dialogues*. arXiv:2601.13206, 2026.